

LAB 7: Doubly Linked list

Insert at Beginning:

```
main.c    Share  Run

1  #include<stdio.h>
2  #include<stdlib.h>
3  struct Node{
4      int data;
5      struct Node *next;
6      struct Node *prev;
7  };
8  struct Node *head=NULL;
9  void insertBeginning(int value){
10     struct Node *newNode=(struct Node*)malloc(sizeof(struct Node));
11     newNode->data=value;
12     newNode->prev=NULL;
13     newNode->next=head;
14     if(head!=NULL)
15         head->prev=newNode;
16     head=newNode;
17 }
18 void display(){
19     struct Node *temp=head;
20     while(temp!=NULL){
21         printf("%d <-> ",temp->data);
22         temp=temp->next;
23     }
24     printf("NULL\n");
25 }
26 int main(){
27     insertBeginning(10);
28     insertBeginning(20);
29     insertBeginning(30);
30 }
```

Output

10 -> 30 -> NULL

Insert At End:

```
main.c
1  #include<stdio.h>
2  #include<stdlib.h>
3  struct Node{
4      int data;
5      struct Node *next;
6      struct Node *prev;
7  };
8  struct Node *head=NULL;
9  void insertEnd(int value){
10     struct Node *newNode=(struct Node*)malloc(sizeof(struct Node));
11     newNode->data=value;
12     newNode->next=NULL;
13     if(head==NULL){
14         newNode->prev=NULL;
15         head=newNode;
16         return;
17     }
18     struct Node *temp=head;
19     while(temp->next!=NULL)
20         temp=temp->next;
21     temp->next=newNode;
22     newNode->prev=temp;
23 }
24 void display(){
25     struct Node *temp=head;
26     while(temp!=NULL){
27         printf("%d <=> ",temp->data);
28         temp=temp->next;
29     }
```

```
30     printf("NULL\n");
31 }
32 int main(){
33     insertEnd(10);
34     insertEnd(20);
35     insertEnd(30);
36     display();
37 }
```

Output

10 <=> 20 <=> 30 <=> NULL